4th Temecula Valley High School Math Competition

Multiple Choice

December, 2020

INSTRUCTIONS

1. DO NOT OPEN TEST BOOKLET UNTIL INSTRUCTED TO DO SO.
2. This is a twenty-five question multiple choice test. Each question is followed by answers marked A, B, C, D and E. Only one of these is correct.
3. Mark your answer to each problem on the scantron with a #2 pencil. Check the blackened circles for accuracy and erase errors and stray marks completely. Only answers properly marked on the answer form will be graded.
4. SCORING: You will receive $+4$ points for each correct answer, -1 point for each problem answered incorrectly, and $+0$ points for each answer left blank.
5. No aids are permitted other than scratch paper, graph paper, rulers, compass, protractors, and erasers. No calculators, smartwatches, or online resources are allowed. No problems on the test will require the use of a calculator.
6. Figures are not necessarily drawn to scale.
7. Before beginning the test, your proctor will ask you to record certain information on the answer form.
8. When your proctor gives the signal, begin working on the problems. You will have 75 minutes to complete the test.

1. Let $a_1 = 1$, $a_2 = 2$, and for $n > 2$ let a_n be the smallest integer greater than a_{n-1} that is the sum of two distinct previous terms in the sequence. What is a_{15}?

   (A) 28 (B) 36 (C) 38 (D) 47 (E) 48

2. What is the sum of the roots of the polynomial $13x^2 - 546x + 780$?

   (A) -60 (B) 60 (C) 21 (D) 42 (E) -42

3. In a triangle ABC, the bisector of $\angle A$ intersects BC at a point D such that $BD = 2$, $DC = 3$, and $AB + AC = 10$. What is AC?
   (A) 6 (B) 4 (C) $\sqrt{3}$ (D) $\frac{\sqrt{3}}{2}$ (E) $2\sqrt{2}$

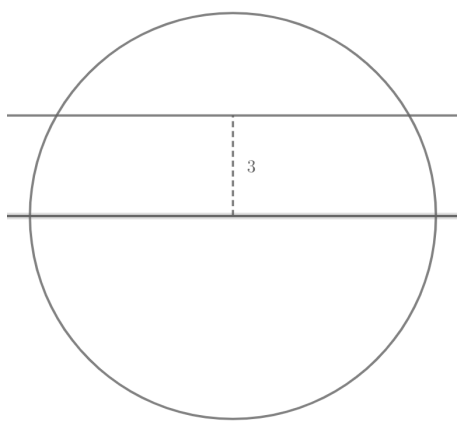
4. Erwin has two bags full of cats: the first has 3 orange cats and 4 black cats, and the second has 2 orange cats and 3 black cats. Erwin picks a bag at random then lets one cat out of the bag at random. What is the probability that he chose the first bag, given that the cat he let out was orange?

   (A) $\frac{15}{29}$ (B) $\frac{8}{29}$ (C) $\frac{29}{35}$ (D) $\frac{4}{7}$ (E) $\frac{2}{7}$

5. Alice wants to send a secret number to Bob. To keep it secret, she first converts it to binary, then flips all the digits (0s become 1s and 1s become 0s). After discarding leading zeros, she converts it back to base 10 and sends it to Bob. Bob receives the number 21. Which of the following could have been Alice's original number?
   (A) 15 (B) 34 (C) 42 (D) 58 (E) 59

6. Carl got Leonhard a circular cake with a radius of 6 inches for his 313th birthday. To Leonhard's dismay, Carl cuts the first slice of the cake by making one cut along the diameter, and another parallel cut 3 inches away from the first cut, as shown below. What is the area of the slice?

   (A) 34 (B) $3\sqrt{3} + 6\pi$ (C) $15\sqrt{3} + 6\pi$ (D) $9\sqrt{3} + 6\pi$ (E) $9\sqrt{3} + 3\pi$



7. The numbers $\frac{10}{1}, \frac{10}{2}, \ldots, \frac{10}{21}$ are written on a blackboard. Each minute, two of the numbers n and m are erased and replaced with $2nm - n - m + 1$. After 20 minutes, there is one number left. What is the sum of all possible values of this number?

   (A) $\frac{1}{2}$ (B) $\frac{3}{10}$ (C) 10 (D) 2 (E) 1

8. What is the units digit of 7^{2021}?

   (A) 1 (B) 7 (C) 9 (D) 3 (E) 4

9. A company makes spherical cows with radius r for physics students to study. They ship 4 spherical cows inside a package whose shape is a regular tetrahedron with side length 1. What is the maximum possible value of r? (Spherical cows are perfectly rigid)

   (A) $\frac{\sqrt{6}-2}{5}$ (B) $\sqrt{2}-1$ (C) $\sqrt{2}$ (D) $\frac{2\sqrt{3}-1}{2}$ (E) $\frac{\sqrt{6}-1}{10}$

10. The company from the previous problem wants to decorate their packaging by coloring each face of the regular tetrahedron either white or black. How many ways can they decorate a package, if two packages are considered the same if one is a rotation of the other?
    (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

11. Let $S_1, S_2, \ldots, S_{2020}$ be nonempty and distinct subsets of $\{1, 2, \ldots, 1010\}$. Also define $S_{2021} = S_1$. Find the maximum value of
$$\sum_{k=1}^{2020} \frac{|S_k \cap S_{k+1}|}{|S_k| \cdot |S_{k+1}|}.$$
    $|S|$ is the number of elements in the set S and $S \cap T$ is the intersection of sets S and T.

    (A) 21 (B) 42 (C) 1011 (D) 1010 (E) 2020

12. Isaac wants to buy exactly n apples, but apples are only sold in bags of $13, 17$ or 23. What is the largest value of n where Isaac cannot buy exactly n apples?

    (A) 73 (B) 77 (C) 84 (D) 92 (E) 96

13. We say an integer is *rising* if its digits form a non-decreasing sequence from left to right. For example, 1337 is rising. How many positive integers with 5 digits are rising?

    (A) 503 (B) 504 (C) 1223 (D) 1260 (E) 1287

14. Lady Lovelace has a Mysterious Machine that takes a positive integer n as input. At each step, It outputs $n/2$ if n is even, and outputs $n+3$ otherwise. The machine then takes this output and uses it as a new input. It stops running when the same input is used twice. What input should Lady Lovelace choose to maximize the number of steps before the machine stops?

    (A) 88 (B) 70 (C) 67 (D) 38 (E) 22

15. We call an integer n a *scrambled square* if there is a permutation $a_1, a_2, \ldots, a_n$ of $\{1, 2, \ldots, n\}$ such that $j+a_j$ is a perfect square for all $1 \leq j \leq n$. What are all the scrambled squares in the set $\{11, 13, 15, 17\}$?

    (A) $11, 13, 15,$ and 17 (B) $13, 15,$ and 17 (C) $11, 13,$ and 15 (D) 13 and 15 (E) None

16. Sophie is making necklaces with 7 beads, each of which is red, green, or blue. How many different necklaces can Sophie make? (Two necklaces are the same if one is a rotation of the other).

    (A) 2187 (B) 2184 (C) 343 (D) 312 (E) 315

17. How many ordered pairs of positive integers (n, m) are there such that $nm^2 + 7$ divides $n^2m + n$ and $n < 2021$?

    (A) 291 (B) 292 (C) 201 (D) 4 (E) 5

18. Abel the Ant stands on a vertex of a cube. At each step, Abel chooses one of the three edges incident to the vertex randomly with equal probability and travels along that edge to the next vertex. What is the probability that Abel visits every vertex of the cube exactly once after 7 steps?

(A) $\frac{2}{243}$ (B) $\frac{2}{729}$ (C) $\frac{7}{243}$ (D) $\frac{1}{8}$ (E) $\frac{1}{3}$

19. What is the value of $\tan(\pi/9) + 4\sin(\pi/9)$?

(A) 1 (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) $\frac{1}{\sqrt{6}-1}$ (E) $\sqrt{6}$

20. Cayley and Hamilton are playing tennis and the score is "deuce," which means that they keep playing until one player has scored two more points than the other. If Cayley has a 2/3 chance of winning each round, what is the probability that Cayley wins the game?

(A) $\frac{4}{5}$ (B) $\frac{2}{3}$ (C) $\frac{4}{9}$ (D) $\frac{5}{9}$ (E) 1

21. A point P is chosen inside a sphere of radius 21 such that the distance from P to the center of the sphere is 20. Two perpendicular chords (line segments intersecting the sphere twice) are drawn through P. What is the sum of the squares of the lengths of the chords?

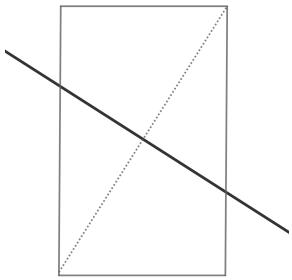
(A) 1916 (B) 1921 (C) 1928 (D) 1963 (E) 1999

22. Emmy writes down a sequence of numbers starting with $a_1 = 12$. She then chooses a_{n+1} from the positive divisors of a_n (including 1 and a_n). She stops the sequence when she first writes down 1. What is the expected value of the length of Emmy's sequence?

(A) $\frac{81}{20}$ (B) $\frac{121}{30}$ (C) $\frac{151}{30}$ (D) 4 (E) 5

23. A new size of rectangular paper, called *golden paper*, has side lengths whose ratio is $\varphi = \frac{1+\sqrt{5}}{2}$. A piece of golden paper is folded along the bold line shown below so that two opposite corners coincide. The area of the resulting 2D shape can be written as $x + \varphi y$. What is $\frac{y}{x}$?

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) 1 (E) 2



24. Find the smallest positive integer with the following property: for all positive integers $m < 2020$ there exists an integer j such that

$$\frac{m}{2020} < \frac{j}{n} < \frac{m}{2021}.$$

(A) 4041 (B) 4040 (C) 3040 (D) 3041 (E) 2021

25. A subset S of $\{1, 2, \ldots, 10\}$ is called *self-aware* if the integer $|S|$ is an element of S ($|S|$ is the number of elements in S). How many self-aware subsets of $\{1, 2, \ldots, 10\}$ are there that do not contain any self-aware subsets?

(A) 89 (B) 84 (C) 55 (D) 50 (E) 49